

Hewitt Tusiime

Dallas, TX — +1 (641) 233-2632 — tusiimehewitt@gmail.com
linkedin.com/in/hewitt-tusiime

Professional Summary

Backend Software Engineer with 5+ years of experience building scalable Java/Spring Boot microservices, distributed systems, and cloud-native applications. Strong in Java 8/11, Spring Boot (Web, Data, Security), REST APIs, Kafka, CI/CD automation, and AWS deployment patterns. Skilled in designing relational databases, optimizing high-volume data workflows, and implementing secure, high-performance backend services. Experienced in leading cross-functional work, mentoring junior engineers, and delivering end-to-end solutions from design through monitoring. Background in ML systems adds strong analytical and data-driven problem-solving capabilities.

Technical Skills

Languages: Java (advanced), Python (advanced), SQL (advanced), JavaScript, R, TypeScript

Frameworks & Tools: Spring Boot, Spring MVC, Spring WebFlux, Spring Security, JPA, Hibernate, JUnit, Mockito

Web: REST APIs, GraphQL, React.js

Cloud: AWS (EC2, S3, RDS, Lambda, ECS, VPC, IAM), Azure (basic), Terraform

DevOps: Docker, Kubernetes, GitHub Actions, GitLab CI, Jenkins, Linux, CI/CD

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Cassandra

Messaging & Data: Apache Kafka, Kafka Streams, Airflow, Hadoop, Spark, dbt

Other: Swagger/OpenAPI, Grafana, Tableau, Jupyter, OOP, Design Patterns (Factory, Strategy, Observer, Singleton, MVC)

Professional Experience

Software Engineer

Makerere University Centre for AI & Data Science, Kampala, Uganda

Aug 2019 – Jun 2024

- Developed and maintained 12+ Spring Boot microservices, exposing REST and GraphQL APIs used by 20k+ monthly users in agriculture and healthcare platforms.
- Optimized PostgreSQL/MySQL data models, improving query performance and reducing ingestion latency by 30% through index tuning and efficient JPA mappings.
- Implemented Kafka Streams pipelines that processed 150k+ messages/day, ensuring scalable and fault-tolerant real-time data ingestion.
- Strengthened backend security using Spring Security + JWT, reducing unauthorized access events by 90%+ with role-based controls.
- Built shared modules for logging, exception handling, and DTO mapping, reducing repetitive code by 25–30% across services.
- Containerized services with Docker and automated blue-green deployments using GitHub Actions + AWS ECS, delivering zero-downtime releases.
- Added Redis caching, retry policies, and circuit breakers, improving API response times by up to 40% during peak loads.
- Converted Python NLP components into Java, improving inference speed by 20% and adding multilingual model support.
- Integrated Spring Actuator and Grafana dashboards, cutting mean-time-to-diagnose (MTTD) by 50% across microservices.
- Led and mentored a team of 4 junior engineers, guiding clean coding practices, cloud skills, testing, and task planning.
- Authored architecture diagrams and API documentation, reducing onboarding time for new team members by 35%.

Key Projects

AI Ticket Classifier – Spring Boot, Python, FastAPI, Hugging Face, Kafka, PostgreSQL

Built a full-stack AI solution where support tickets are automatically classified using text embeddings and transformer-based language models. Java microservices handled ticket routing and authentication, FastAPI served ML inference, Kafka enabled asynchronous messaging, and PostgreSQL stored predictions for analytics dashboards.

Conference Management System – Java, Spring Boot, MySQL, AWS

Developed backend APIs for submissions, reviews, and payments. Containerized services, implemented secure access layers, and deployed using AWS ECS + Terraform.

Veterinary Clinic System – Java, Spring Boot, MySQL

Built appointment scheduling and patient-record modules with Spring Boot, designed relational schemas, and automated deployments using CI/CD pipelines.

Publications

AI Letters (2024): Building Text and Speech Benchmark Datasets for East African Languages. DOI: 10.1002/ail2.92

Makerere AI Lab (2021): Misinformation Detection in Luganda-English Code-Mixed Social Media Text

Education

M.S., Computer Science, Maharishi International University, USA

Expected 2027

B.S., Software Engineering, Makerere University, Uganda

2022